

PENETRATION TESTING REPORT

NETWORK SCANNING WITH ZENMAP

W2-PM5 | WEEK 2 | PROJECT MODULE 5 | NETWORKWALKS

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Program / Batch	Cybersecurity & Ethical Hacking — Networkwalks
Date	18 September 2026
Module completed	W2-PM5 (Network Scanning with Zenmap)
Client / Target	My own local LAN network (192.168.0.0/24)
Permission secured from client?	Yes — the network and all devices scanned are my own
Phase covered	Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on the network and devices that I own myself. All of this material is for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers Project Module 5 of Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks: network scanning of my own local area network using Zenmap, the official GUI front-end for Nmap.

The objective was to install Zenmap on a Windows PC, identify my own IP address and LAN subnet, discover the live hosts on that subnet, record their IP and MAC addresses, and save the resulting network topology as a PDF. This is the scanning and network-discovery phase of a penetration test, which normally follows the reconnaissance phase.

Each task below is presented with the exact command used, the result observed, the screenshot taken while performing that step, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this module and its purpose.

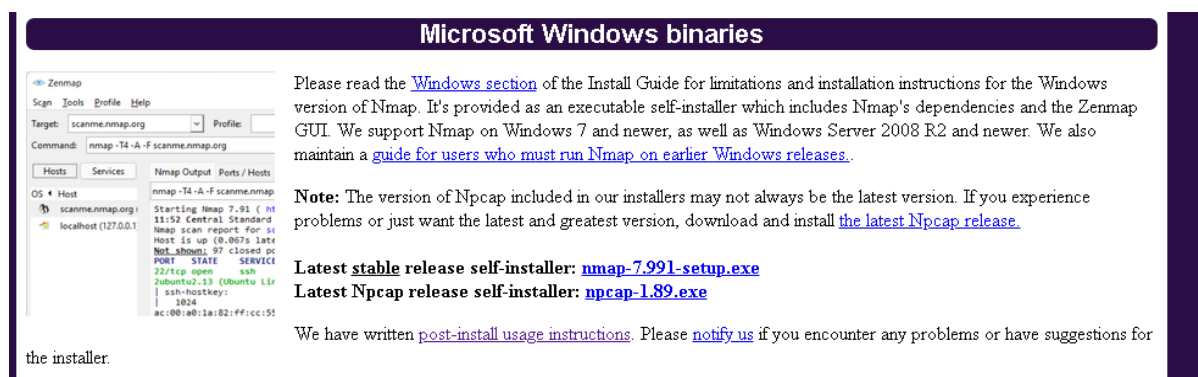
Tool	Purpose
Windows 11 PC	Host operating system used to perform the scanning activity
Nmap 7.991	Network scanning engine installed from the official nmap.org site
Zenmap (Nmap GUI)	Graphical front-end used to run the ping scan and draw the topology
Npcap	Packet capture driver required by Nmap on Windows
Windows CMD (ipconfig)	Identify the local IP address, subnet mask and default gateway

4. Activities Performed

Task 1 — Download & install Zenmap from the official website

I downloaded Nmap from the official project website, <https://nmap.org/download.html>, and used the latest stable Windows self-installer, `nmap-7.991-setup.exe`. This installer bundles the Nmap engine, the Npcap packet capture driver and the Zenmap graphical front-end.

During setup I kept the default components (Nmap Core Files, Npcap, Zenmap GUI, Ndiff, Ncat and Nping) and the default installation folder, then completed the wizard and launched Zenmap from the desktop shortcut.



Screenshot 1 — Official nmap.org download page showing the Windows self-installer used

Why it matters: Security tools must always be taken from the official source. Nmap installers repackaged on third-party download sites are a known malware delivery route, so verifying the source is itself a security control.

Task 2 — Find my local IP address & LAN subnet

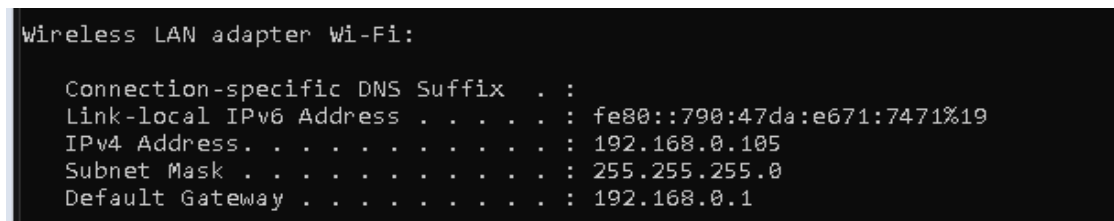
I opened Windows Command Prompt and ran:

```
ipconfig
```

The output for the Wireless LAN adapter Wi-Fi gave the following configuration:

- IPv4 Address: 192.168.0.105
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.0.1

A subnet mask of 255.255.255.0 is a /24 prefix, so my LAN subnet is 192.168.0.0/24, giving a usable host range of 192.168.0.1 to 192.168.0.254 (254 possible hosts).



Screenshot 2 — `ipconfig` output showing the local IPv4 address, subnet mask and default gateway

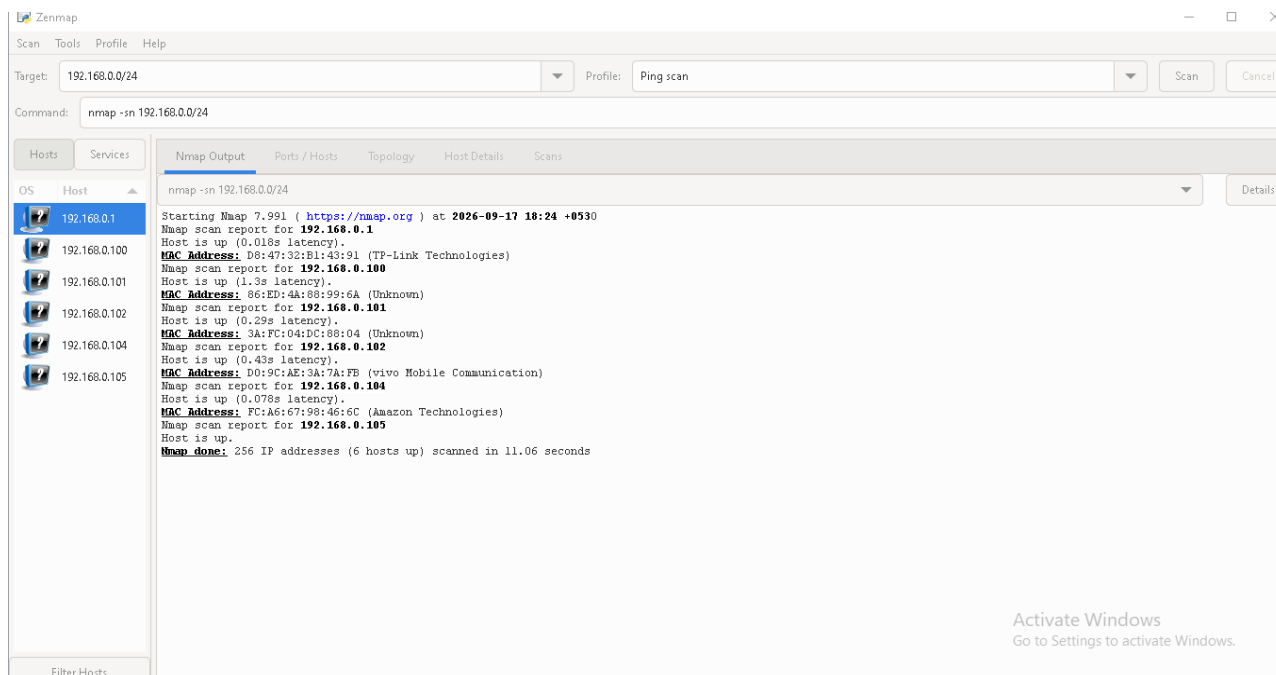
Why it matters: The subnet mask tells an attacker exactly how large the target network is and which address range to sweep. The default gateway is almost always the router, which is the highest-value device on a home network.

Task 3 — Find the list of live hosts in my IP subnet

I opened Zenmap, entered 192.168.0.0/24 in the Target field and selected the Ping scan profile. Zenmap built the command:

```
nmap -sn 192.168.0.0/24
```

The `-sn` flag performs host discovery only — it sends ARP/ICMP probes to find live hosts and skips port scanning entirely, which makes it fast and comparatively quiet. Nmap scanned all 256 addresses in 11.06 seconds.



Screenshot 3 — Zenmap ping scan of 192.168.0.0/24 listing every live host with its MAC address

Why it matters: This single command turns an unknown network into a target list in a few seconds. It is normally the first action an attacker takes after gaining access to a Wi-Fi network.

Task 4 — How many hosts are live in my subnet?

Answer: 6 hosts are live (including my own PC). Nmap reported: 256 IP addresses (6 hosts up) scanned in 11.06 seconds.

Task 5 & Task 6 — IP addresses and MAC addresses of the live hosts

The ping scan in Screenshot 3 returned the following hosts. Nmap does not display a MAC address for the scanning machine itself, because that host is not reached over the network.

#	IP Address	MAC Address	Vendor / Identification
1	192.168.0.1	D8:47:32:B1:43:91	TP-Link Technologies — default gateway / router
2	192.168.0.100	86:ED:4A:88:99:6A	Unknown (randomised MAC — likely a mobile device)
3	192.168.0.101	3A:FC:04:DC:88:04	Unknown (randomised MAC — likely a mobile device)
4	192.168.0.102	D0:9C:AE:3A:7A:FB	vivo Mobile Communication — smartphone
5	192.168.0.104	FC:A6:67:98:46:6C	Amazon Technologies — smart/IoT device
6	192.168.0.105	Not shown by Nmap (local host)	My own Windows PC — use ipconfig /all for its MAC

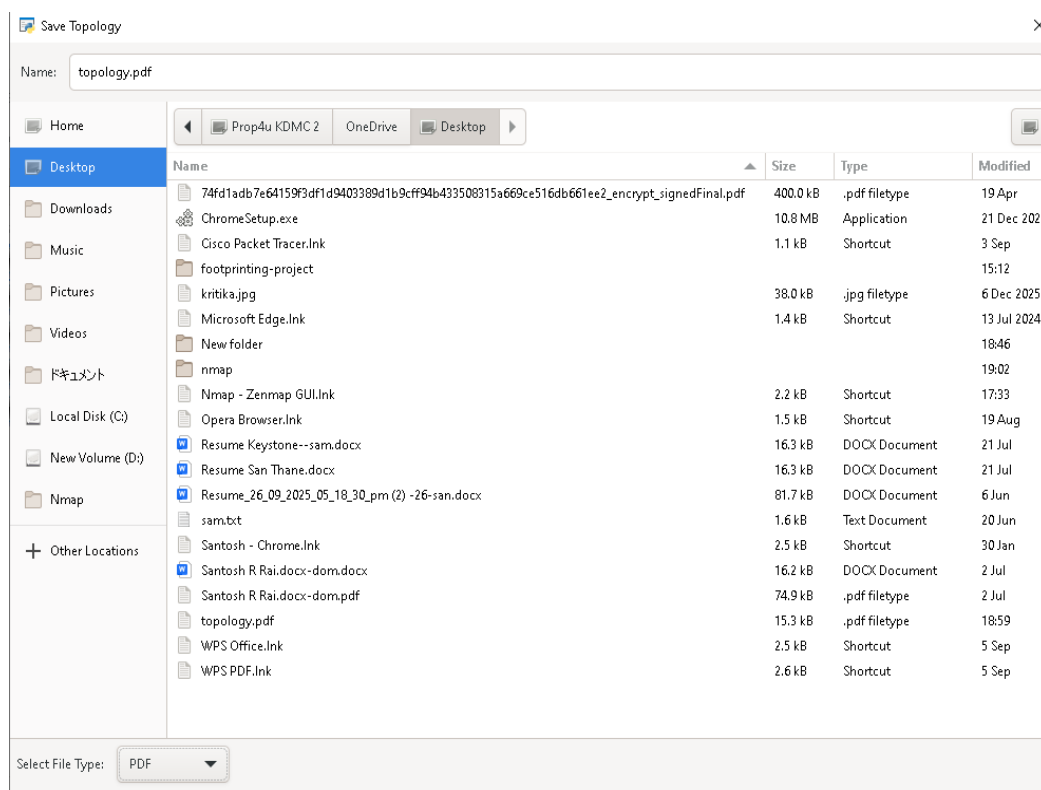
Note: 192.168.0.100 and 192.168.0.101 show vendors as "Unknown" because their MAC addresses are locally administered (randomised privacy MACs used by modern phones), so the OUI prefix does not map to a registered manufacturer.

Why it matters: MAC addresses reveal the manufacturer of each device through the OUI prefix. Knowing that a network contains a TP-Link router, a vivo phone and an Amazon device lets an attacker research the known weaknesses of those specific products before touching them.

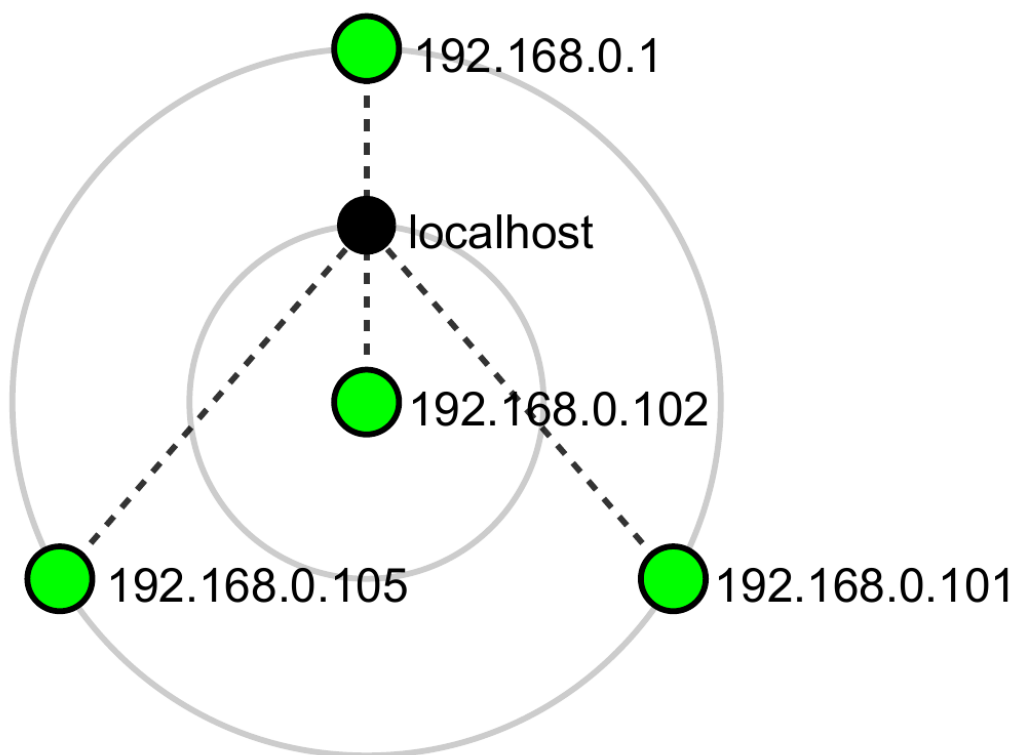
Task 7 — Display & save the output topology in PDF format

After the scan finished I opened the Topology tab in Zenmap, turned on the Legend and reviewed it. Green nodes indicate hosts that were not port-scanned, the black node in the centre is localhost (my own PC), and the dashed lines show that no traceroute information was collected — expected, since every host is a single hop away on the same LAN.

I then clicked Save Graphic, selected PDF as the file type and saved the file as topology.pdf on my desktop.



Screenshot 4 — Save Topology dialog with PDF selected as the output format, saving to the desktop



Screenshot 5 — The saved topology.pdf showing the discovered hosts around localhost

Why it matters: A topology diagram converts a list of addresses into a picture of the network. Defenders use it to document what should be present; an attacker uses the same picture to choose a pivot point.

5. Risk Analysis / Impact

Based on the information collected during this scanning activity, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Host discovery possible from inside the LAN	A single ping scan mapped 6 live hosts in about 11 seconds	Any device that joins the Wi-Fi can enumerate the whole network	● Medium
2	Devices respond to ICMP echo requests	All hosts replied to the -sn probe	Makes every device easy to locate and target for follow-up scanning	● Low
3	Device vendors exposed through MAC OUIs	TP-Link, vivo and Amazon identified from MAC prefixes	Reveals device types, helping an attacker pick a weak target	● Medium
4	Router reachable at a predictable address	Gateway found at 192.168.0.1	Router admin page becomes an obvious target for default-credential attacks	● Medium
5	Unidentified hosts present on the network	192.168.0.100 and .101 returned unknown vendors	An unauthorised or forgotten device could go unnoticed	● Medium

Risk level key: ● Critical ● Medium ● Low

The items above are observations from a host-discovery exercise, not confirmed vulnerabilities. This module involved discovery only — no port scanning, service enumeration, exploitation or vulnerability validation was

performed. The presence of a device on the network does not by itself mean that the device is vulnerable, and further authorised testing would be required to confirm any actual weakness.

6. Recommendations

Based on the observations from this activity, I recommend the following security improvements:

1. **Maintain an inventory of network devices** — Every device on the LAN should be known and accounted for, so that anything unexpected stands out immediately.
2. **Investigate unknown hosts** — The two hosts with unidentified vendors should be traced back to a known owner and device, or removed from the network.
3. **Secure the router** — Change the default administrator password on the TP-Link gateway, disable remote management and keep its firmware updated.
4. **Use strong Wi-Fi security** — Enable WPA2/WPA3 with a long unique passphrase, since everything in this report was possible only from inside the wireless network.
5. **Separate IoT and guest devices** — Smart devices and visitors should sit on a guest network or VLAN so they cannot enumerate or reach personal computers.
6. **Enable host firewalls** — Windows Defender Firewall should remain on so that devices do not respond to unnecessary probes from other hosts.
7. **Perform regular internal scans** — Repeat this ping scan periodically and compare results to spot new or unauthorised devices.
8. **Keep network documentation updated** — Save the topology output with each scan so that changes to the network can be tracked over time.
9. **Scan only with authorisation** — Network discovery should only ever be performed against networks you own or have written permission to test.

7. Conclusion

In this module I installed Nmap and Zenmap on a Windows PC and used them to map my own home network. Starting from nothing more than the output of ipconfig, I identified my subnet as 192.168.0.0/24, ran a ping scan against all 256 addresses and discovered 6 live hosts, together with the MAC address and manufacturer of each one, in about eleven seconds.

I learned that the `-sn` option performs host discovery without port scanning, that a subnet mask directly defines the scope of a scan, and that MAC OUI prefixes identify device manufacturers — while randomised MAC addresses on modern phones deliberately break that identification. I also learned to read the Zenmap topology view and its legend, and to export it as evidence for a report.

More broadly, this exercise showed how much an attacker gains from simply being on the same network. A device list, a router address and a set of device types are all available before a single exploit is attempted, which is exactly why segmentation, strong Wi-Fi credentials and device inventories matter.

Finally, all scanning was carried out inside my own network as part of an assigned educational lab, which is the only context in which these tools should be used.

- End -

Author

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Project Information

Program: Cybersecurity program at Networkwalks | Week: 02 | Module: W2-PM5 — Network Scanning with Zenmap